

## CROKER RIVER, NU, Canada

WMO: 710590

Lat: 69.280N

Lon: 119.220W

Elev: 228

StdP: 14.58

Time Zone: -7.00 (NAM)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -32.6      | -30.0 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 46.2                  | -2.7 | 41.9 | -2.5 | 12.9                  | 120  | 1.087 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 7             | 12.6                   | 70.6            | N/A  | 65.6 | N/A  | 61.1 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 13.1                 | 100  |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |     |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|-----|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |     |
| Min               |      | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        | Min  | Max |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |
| 38.0              | 32.2 | 28.0 |                            | -35.8 | 77.1 | 3.0                | 4.6                                                | -38.0     | 80.4 | -39.8      | 83.1 | -41.5      | 85.7 | -43.7      | 89.0 |     |
|                   |      |      |                            | N/A   | N/A  | N/A                | N/A                                                | N/A       | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A |
|                   |      |      |                            | DB    |      |                    |                                                    |           |      |            |      |            |      |            |      |     |
|                   |      |      |                            | WB    |      |                    |                                                    |           |      |            |      |            |      |            |      |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 15.1   | -11.7 | -13.4 | -10.4 | 3.7   | 22.5  | 41.2  | 48.6  | 46.5  | 36.2  | 20.2  | 4.1   | -7.8  |
|                                                                        | DBStd       | 24.74  | 11.04 | 10.71 | 10.76 | 10.84 | 10.17 | 8.52  | 8.31  | 7.75  | 7.00  | 8.06  | 9.27  | 10.03 |
|                                                                        | HDD50       | 12900  | 1913  | 1776  | 1874  | 1390  | 851   | 293   | 129   | 165   | 418   | 923   | 1376  | 1792  |
|                                                                        | HDD65       | 18208  | 2378  | 2196  | 2339  | 1840  | 1316  | 715   | 512   | 576   | 865   | 1388  | 1826  | 2257  |
|                                                                        | CDD50       | 173    | 0     | 0     | 0     | 0     | 0     | 27    | 87    | 56    | 3     | 0     | 0     | 0     |
|                                                                        | CDD65       | 6      | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 2     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 56     | 0     | 0     | 0     | 0     | 0     | 13    | 25    | 18    | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 4      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 13.2   | 14.7  | 13.0  | 12.6  | 12.4  | 12.5  | 13.0  | 12.2  | 12.1  | 13.2  | 14.0  | 14.4  | 13.9  |
|                                                                        | PrecAvg     | 8.6    | 0.4   | 0.3   | 0.4   | 0.3   | 0.4   | 0.7   | 1.5   | 1.7   | 1.2   | 0.8   | 0.6   | 0.4   |
| (15) Precipitation                                                     | PrecMax     | 10.7   | 1.0   | 1.1   | 1.5   | 0.7   | 1.2   | 1.5   | 3.5   | 3.7   | 2.6   | 1.5   | 1.6   | 1.1   |
|                                                                        | PrecMin     | 5.7    | 0.1   | 0.1   | 0.1   | 0.0   | 0.1   | 0.2   | 0.5   | 0.8   | 0.3   | 0.3   | 0.2   | 0.1   |
|                                                                        | PrecStd     | 1.4    | 0.2   | 0.2   | 0.3   | 0.2   | 0.3   | 0.3   | 0.6   | 0.7   | 0.5   | 0.3   | 0.3   | 0.2   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 24.9  | 19.2  | 23.4  | 37.3  | 49.3  | 75.5  | 77.8  | 76.5  | 59.3  | 41.6  | 31.7  | 22.1  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | DB     | 15.7  | 14.4  | 16.0  | 29.0  | 43.4  | 66.5  | 71.1  | 69.3  | 53.5  | 35.2  | 25.5  | 15.4  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | DB     | 9.8   | 8.3   | 10.3  | 23.8  | 39.4  | 60.3  | 66.4  | 63.5  | 49.5  | 32.5  | 21.3  | 11.1  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | DB     | 3.7   | 2.2   | 5.6   | 19.4  | 35.7  | 55.1  | 62.3  | 58.7  | 45.8  | 30.5  | 17.7  | 6.8   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.1  | 11.2  | 11.5  | 12.1  | 10.1  | 11.7  | 12.6  | 11.0  | 7.7   | 8.3   | 9.9   | 10.0  |
|                                                                        |             | MCDBR  | 17.4  | 16.9  | 16.5  | 15.5  | 14.4  | 23.3  | 21.5  | 20.1  | 14.6  | 8.4   | 12.2  | 13.5  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.158 | 0.207 | 0.246 | 0.292 | 0.332 | 0.328 | 0.355 | 0.339 | 0.303 | 0.232 | 0.155 | N/A   |
|                                                                        | taud        |        | 1.887 | 2.049 | 2.188 | 2.127 | 2.131 | 2.357 | 2.382 | 2.480 | 2.491 | 2.170 | 1.919 | N/A   |
|                                                                        | Ebn at Noon |        | 16    | 180   | 250   | 267   | 268   | 275   | 262   | 254   | 233   | 165   | 14    | 0     |
|                                                                        | Edh at Noon |        | 1     | 13    | 23    | 34    | 39    | 32    | 30    | 25    | 18    | 11    | 1     | 0     |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6     | 147   | 671   | 1457  | 2000  | 2007  | 1685  | 1062  | 508   | 140   | 18    | 0     |
|                                                                        | RadStd      |        | 1     | 14    | 31    | 68    | 160   | 164   | 129   | 108   | 56    | 19    | 2     | 0     |

## Historical Trends

|                             | DBAvg | Heating |       |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |       | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)   | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (6 neighbors) | N/A   | +2.47   | +2.88 | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page